1 **(a)** 5 **[B1]**

(b) $\dfrac{1}{36}$ **[B1]**

(c) t^8 **[B1]**

[Total Marks: 3]

2 **(a)** $16^{\frac{1}{4}} = 2$ or equivalent correct first step **[M1]**

$\dfrac{1}{8}$ **[A1]**

(b) 25 **[B1]**

x^6y^{-4} or $\dfrac{x^6}{y^4}$ **[B1]**

(c) Expresses terms with a common base, e.g. $4^x = 2^{2x}$ or $32 = 2^5$ **[M1]**

Sets up a correct linear equation from indices, e.g. $3x - 1 + 2x = 5$ **[M1]**

$x = 1.2$ or $x = \dfrac{6}{5}$ **[A1]**

[Total Marks: 7]

3 **(a)** Inverts fraction to remove negative power or correctly applies negative power to all terms, e.g. $\left(\dfrac{y^4}{16x^8}\right)^{\frac{1}{2}}$ or $\dfrac{16^{-\frac{1}{2}}x^{-4}}{y^{-2}}$ **[M1]**

Correctly evaluates the coefficient $16^{-\frac{1}{2}} = \dfrac{1}{4}$ or correctly halves powers **[M1]**

$\dfrac{y^2}{4x^4}$ or $0.25x^{-4}y^2$ **[A1]**

(b) Recognises that $4^x = 2^{2x}$ or substitutes $u = 4^x$ **[M1]**

Expresses 2^{2x+1} as 2×2^{2x} or $2(4^x)$ **[M1]**

Simplifies to a single exponential term, e.g. $-2^{2x} = -8$ or $u - 2u = -8$ **[M1]**

$x = 1.5$ or $x = \dfrac{3}{2}$ **[A1]**

[Total Marks: 7]